

Reconnaissance & Information Gathering

PASSIVE RECONNAISSANCE

1. whois tryhackme.com

query public databases to retrieve registration and ownership information for internet resources

Field	Data Found
Domain Name	tryhackme.com
Registrar	NameCheap, Inc.
Creation Date	2018-07-05
Expiry Date	2034-07-05
Name Servers	KIP.NS.CLOUDFLARE.COM, UMA.NS.CLOUDFLARE.COM
DNSSEC	unsigned
Domain Status	clientTransferProhibited

2. nslookup tryhackme.com

query Domain Name System (DNS) servers to translate domain names into IP addresses and vice versa

Field	Data Found
A Record (IPv4)	104.20.29.66
A Record (IPv4) Secondary address	172.66.164.239
AAAA Record (IPv6)	2606:4700:83b6:d299:d33b:485:ae35:d89c
Resolved By	127.0.0.53

3. nslookup -type=TXT thmlabs.com

retrieve TXT records associated with a specific domain

Server: 127.0.0.53
Address: 127.0.0.53#53
Non-authoritative answer:
thmlabs.com text = "THM{a5b83929888ed36acb0272971e438d78}"

4. dig tryhackme.com MX

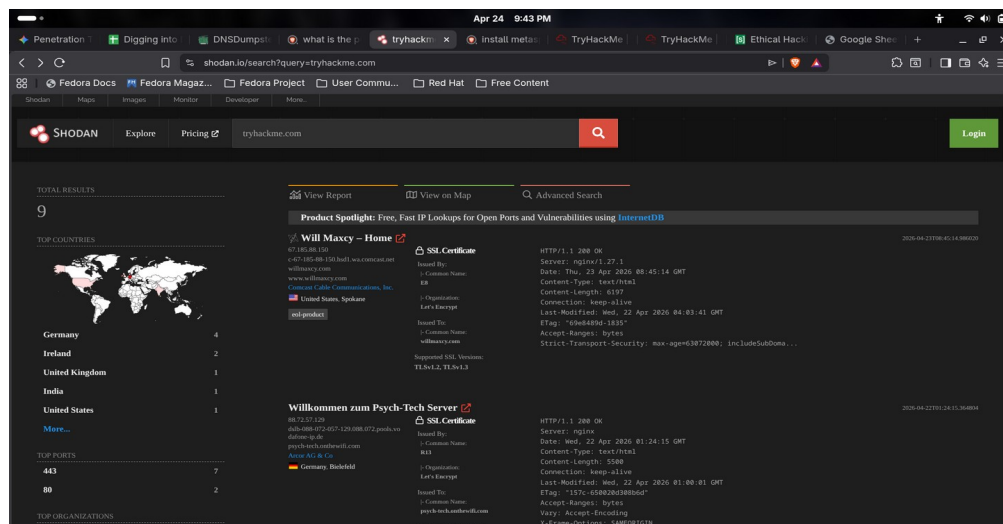
queries the DNS server to retrieve the Mail Exchange (MX) records for the specified domain

Technical Detail	Value
Record Type	MX (Mail Exchanger)
Primary Server	aspmx.l.google.com
Redundancy	5 Servers
TTL	154 Seconds

Status	NOERROR
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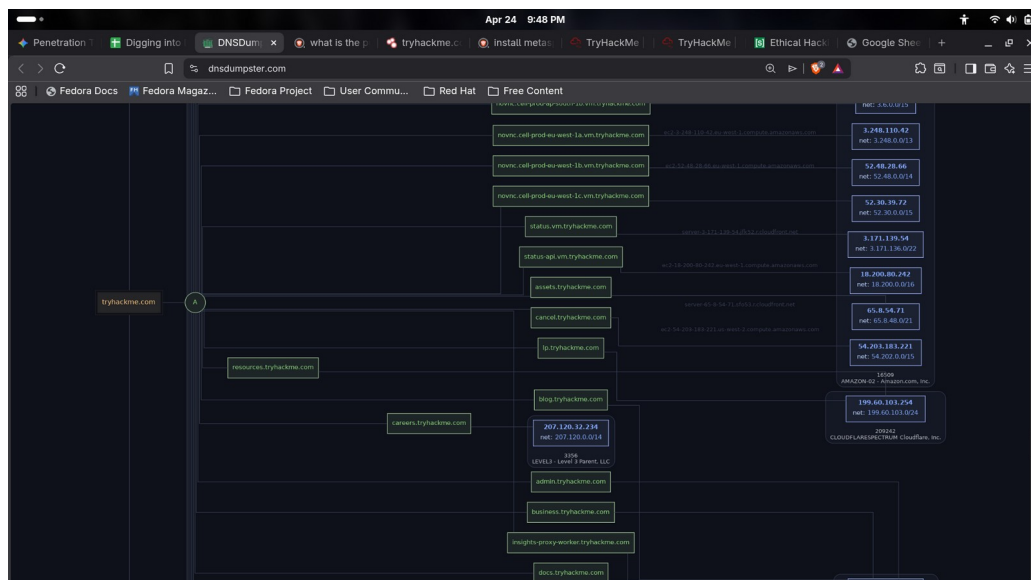
5. Shodan.io

- OSINT tool
- specialized search engine for internet-connected devices
- IP address, hosting company, geographic location, server type and version



6. DNSDumpster

- offers detailed answers to queries
- can get the details of subdomains



ACTIVE RECONNAISSANCE

1. traceroute – 4 scanme.nmap.org

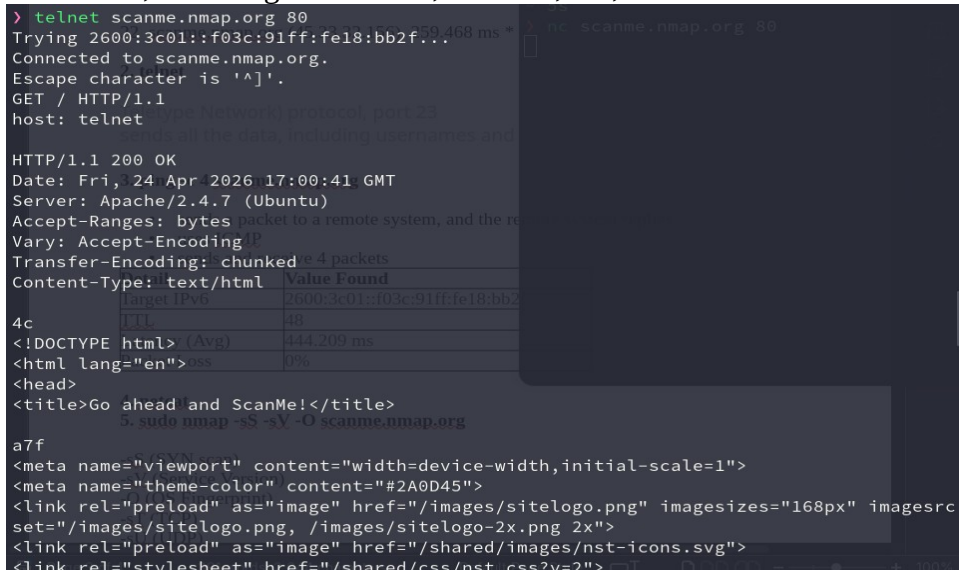
- maps the end-to-end network path by identifying every router
- incrementing TTL
- identified the exact transition point where traffic left the local ISP in India and entered international transit in Los Angeles.
- target is protected by enterprise-tier infrastructure, passing through Lumen and Akamai nodes.

Output

```
1 _gateway (172.17.119.133) 2.488 ms 2.432 ms 2.653 ms
2 * * *
3 255.0.0.2 (255.0.0.2) 203.851 ms 203.820 ms 203.815 ms
....
22 scanme.nmap.org (45.33.32.156) 259.468 ms * *
```

2. telnet scanme.nmap.org 80

- Teletype Network) protocol, port 23
- Sends all the data, including server used, versions, OS, etc



```
> telnet scanme.nmap.org 80
Trying 2600:3c01::f03c:91ff:fe18:bb2f...59.468 ms *
Connected to scanme.nmap.org.
Escape character is '^J'.
GET / HTTP/1.1
host: telnet
HTTP/1.1 200 OK
Date: Fri, 24 Apr 2026 17:00:41 GMT
Server: Apache/2.4.7 (Ubuntu)
Accept-Ranges: bytes
Vary: Accept-Encoding
Transfer-Encoding: chunked
Content-Type: text/html
4c
<!DOCTYPE html> (Avg)
<html lang="en">
<head>
<title>Go ahead and ScanMe!</title>
<meta name="viewport" content="width=device-width,initial-scale=1">
<meta name="theme-color" content="#2A0D45">
<link rel="preload" as="image" href="/images/sitelogo.png" imagesizes="168px" imagesrc
set="/images/sitelogo.png, /images/sitelogo-2x.png 2x">
<link rel="preload" as="image" href="/shared/images/nst-icons.svg">
<link rel="stylesheet" href="/shared/css/nst.css?v=2">
```

3. ping -c 4 scanme.nmap.org

- sends a packet to a remote system, and the remote system replies
- uses ICMP
- sends and receive 4 packets

Detail	Value Found
Target IPv6	2600:3c01::f03c:91ff:fe18:bb2f
TTL	48
Latency (Avg)	444.209 ms
Packet Loss	0%

4. nc scanme.nmap.org 80

- netcat
- function as a client that connects to a listening port,
- support both TCP and UDP
- similar to telnet

-l	Listen mode
-p	Specify the Port number
-n	Numeric only; no resolution of hostnames via DNS

5. **sudo nmap -sS -sV -O scanme.nmap.org**

-sS (SYN scan)
-sV (Service Version)
-O (OS Fingerprint)
-sT (TCP)
-sU (UDP)

- Network Mapper (nmap)
- Free and open-source utility designed for network discovery and security auditing

Field	Finding
IP Address	45.33.32.156
Port 22 (SSH)	OpenSSH 6.6.1p1
Port 80 (HTTP)	Apache 2.4.7
Port 31337	tcpwrapped
Filtered Ports	25, 135, 139, 445
OS Detection	Linux Kernel